

Principal Component Analysis — by hand

Four points, two variables, one pencil. No computer needed.

1 Data & means

	x	y
A	0	1
B	1	2
C	3	3
D	4	2
Σ	8	8

$$\bar{x} = 8 / 4 = 2$$
$$\bar{y} = 8 / 4 = 2$$

3 Eigenvalues

$$\det(S - \lambda I) = 0$$
$$(10 - \lambda)(2 - \lambda) - 3 \cdot 3 = 0$$
$$\lambda^2 - 12\lambda + 11 = 0$$
$$(\lambda - 11)(\lambda - 1) = 0$$

$$\lambda_1 = 11 \quad \lambda_2 = 1$$

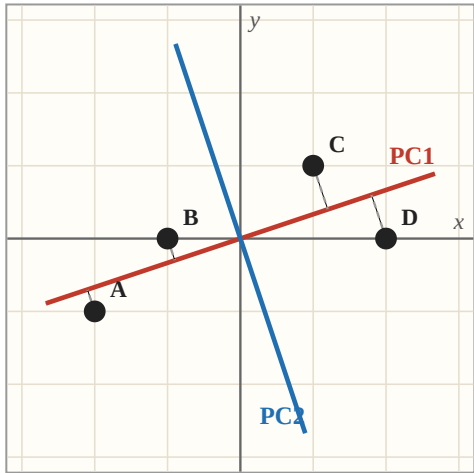
✓ trace $10+2 = 12 = 11+1$ · det $20-9 = 11 = 11 \cdot 1$
(of C: $11/3$ and $1/3$)

4 Eigenvectors

$$\lambda = 11: (S - 11I)v = 0$$
$$-v_1 + 3v_2 = 0 \rightarrow v = (3, 1)$$
$$\text{length } \sqrt{10} \rightarrow \text{PC1} = (3, 1)/\sqrt{10}$$

$$\lambda = 1: 9v_1 + 3v_2 = 0 \rightarrow v = (1, -3)$$
$$\text{PC2} = (1, -3)/\sqrt{10}$$

✓ perpendicular: $3 \cdot 1 + 1 \cdot (-3) = 0$



Centred data with the two principal axes. Dashed: projection onto PC1.

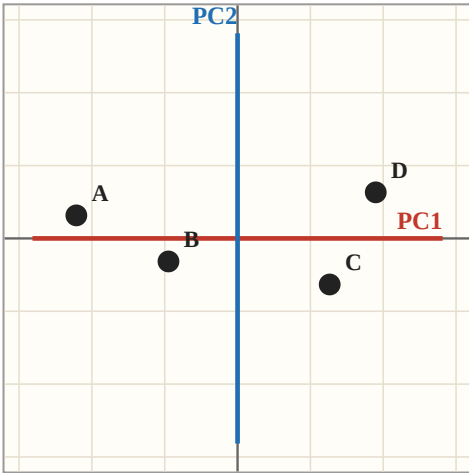
5 Scores (project the data)

$$\text{PC1} = (3a + b)/\sqrt{10}$$

$$\text{PC2} = (a - 3b)/\sqrt{10}$$

obs	PC1	PC2
A	-2.21	0.32
B	-0.95	-0.32
C	1.26	-0.63
D	1.90	0.63
Σ score ²	11	1

✓ Σ score² = eigenvalue: $(49+9+16+36)/10 = 11$ and $(1+1+4+4)/10 = 1$



Same points, rotated into the PC1–PC2 frame.

6 Variance explained

$$\text{PC1: } 11 / (11+1) = 91.7 \% \quad \text{PC2: } 1/12 = 8.3 \%$$



One axis keeps almost all the information: 2-D → 1-D.

A little history. Karl Pearson (1901) posed it geometrically as the “line of closest fit” to points in space; Harold Hotelling (1933) named the “principal components” and computed them with an iterative power method on a desk calculator. Both did precisely what this poster does — sums of squares and products, then a characteristic equation — by hand.

Recipe. centre → covariance → eigen-decomposition → sort by λ → project.
Data: A(0,1) B(1,2) C(3,3) D(4,2) — numbers chosen so the eigenvalues (11, 1) are whole.